

PROJECT BUTTON HR–V0 FOUR–CHANNEL RUNTIME OBSERVATION INTERFACE

DIAGNOSTIC–ONLY ISOLATED RECEIVER CANDIDATE; NO GPIO ALLOCATION; ZERO SAFETY CREDIT

01 Field and compute boundaries

File: 01_boundaries.kicad_sch

02 SR1 and SRA1 Type–3 inputs

File: 02_sr1_sra1_inputs.kicad_sch

03 K1 and K2 diagnostic auxiliary inputs

File: 03_k1_k2_inputs.kicad_sch

04 Fail–low compute outputs

File: 04_compute_outputs.kicad_sch

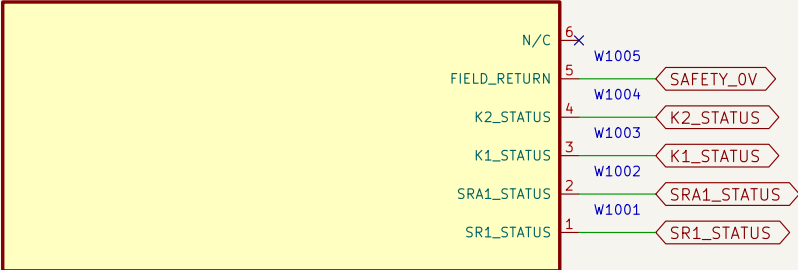
V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /		
File: hr–v0–runtime–observation–interface–p0.1.kicad_sch		
Title: PROJECT BUTTON HR–V0 FOUR–CHANNEL RUNTIME OBSERVATION INTERFACE		
Size: A3	Date: 2026–08–10	Rev: R201 / P0.1
KiCad E.D.A. 10.0.5	Id: 1/5	

01 Field and compute boundaries

Six-position field and logic terminal candidates; no Pi GPIO allocation.

JFIELD1

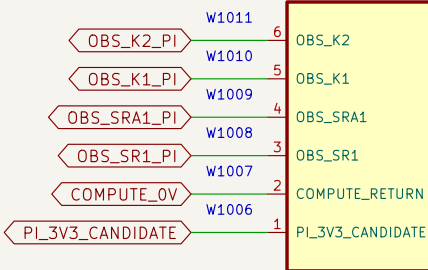
Phoenix MKDS field terminal candidates: 1751264 plus 1751248



EXACT PCB TERMINAL CANDIDATES – MATING/HARNESS/PCB HOLD

JLOGIC1

Phoenix MKDS compute terminal candidates: 1751264 plus 1751248



EXACT PCB TERMINAL CANDIDATES – MATING/HARNESS/PCB HOLD

NOTE 1: Four positive diagnostic states only. No receiver output commands motion.

NOTE 2: SAFETY_OV and COMPUTE_OV remain distinct; no new bond is authorized.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /01 Field and compute boundaries/

File: 01_boundaries.kicad_sch

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BOUNDARY1

GALVANIC DOMAIN BOUNDARY – ZERO SAFETY CREDIT

FIELD_FGND

LOGIC_GND1

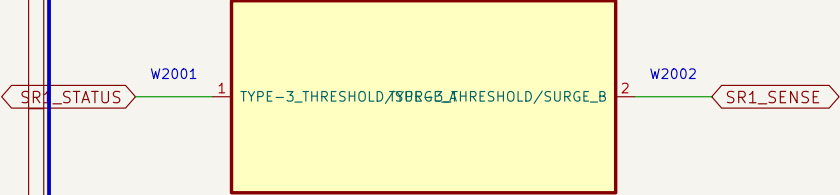
LOGIC

COMPUTE_OV

02 SR1 and SRA1 Type-3 inputs

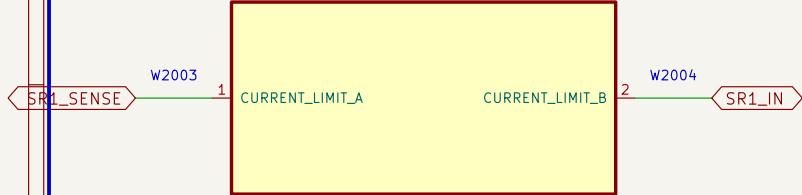
Y32 load budget and existing H1 application remain held.

RTH1
1.00 kohm 1% 0.4 W MELF; Vishay MMA02040C1001FB300



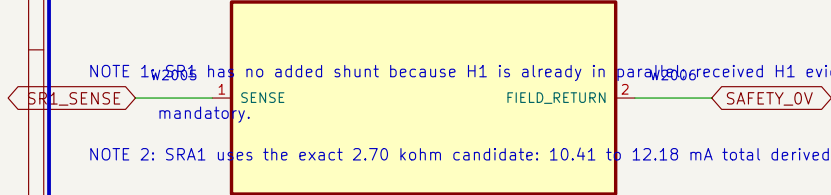
EXACT COMPONENT CANDIDATE – APPLICATION/PCB/PHYSICAL HOLD

RSN1
562 ohm 1% 0.125 W 0805; Panasonic ERJ6ENF5620V



EXACT COMPONENT CANDIDATE – APPLICATION/PCB/PHYSICAL HOLD

CFI1
10 nF 50 V X7R; TDK CGA3E2X7R1H103K080AA

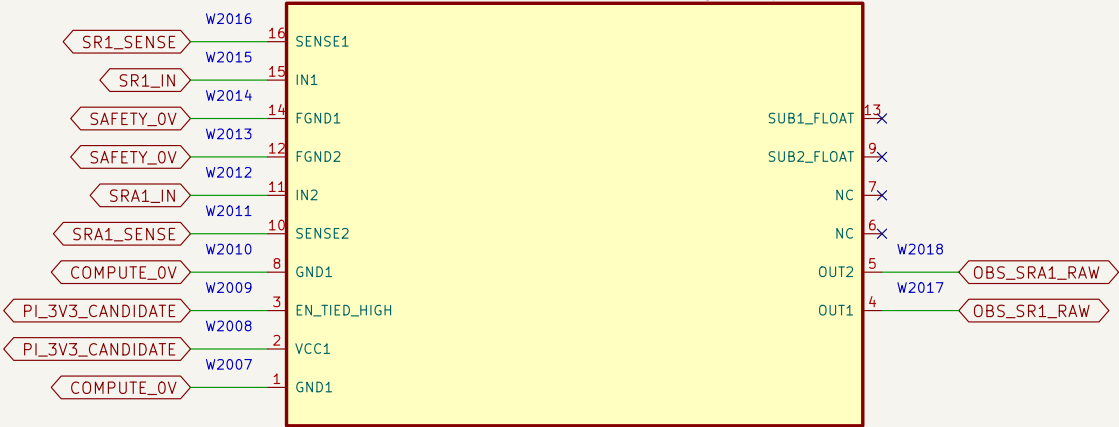


NOTE 1: SR1 has no added shunt because H1 is already in parallel. Received H1 evidence is mandatory.

NOTE 2: SRA1 uses the exact 2.70 kohm candidate: 10.41 to 12.18 mA total derived screen.

EXACT COMPONENT CANDIDATE – DC-BIAS/PCB/EMC HOLD

UOBS1
Texas Instruments ISO1212DBQ dual isolated digital-input receiver



EXACT IC CANDIDATE – PCB/EMC/FAULT/PHYSICAL HOLD

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /02 SR1 and SRA1 Type-3 inputs/

File: 02_sr1_sra1_inputs.kicad_sch

Title: PB HR-V0 R201 / P0.1 – 02

Size: A4

Date: 2026-08-10

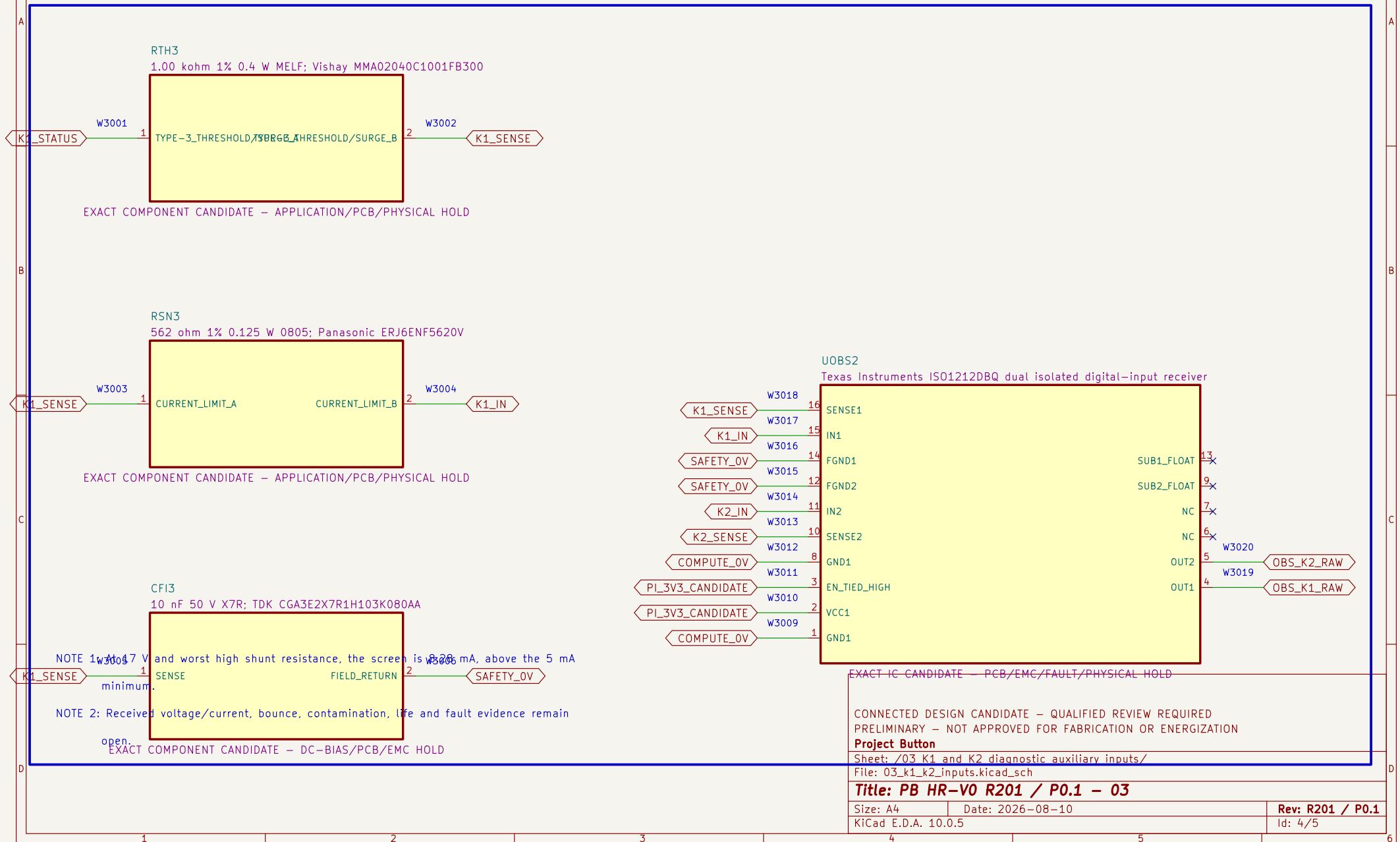
Rev: R201 / P0.1

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Id: 3/5

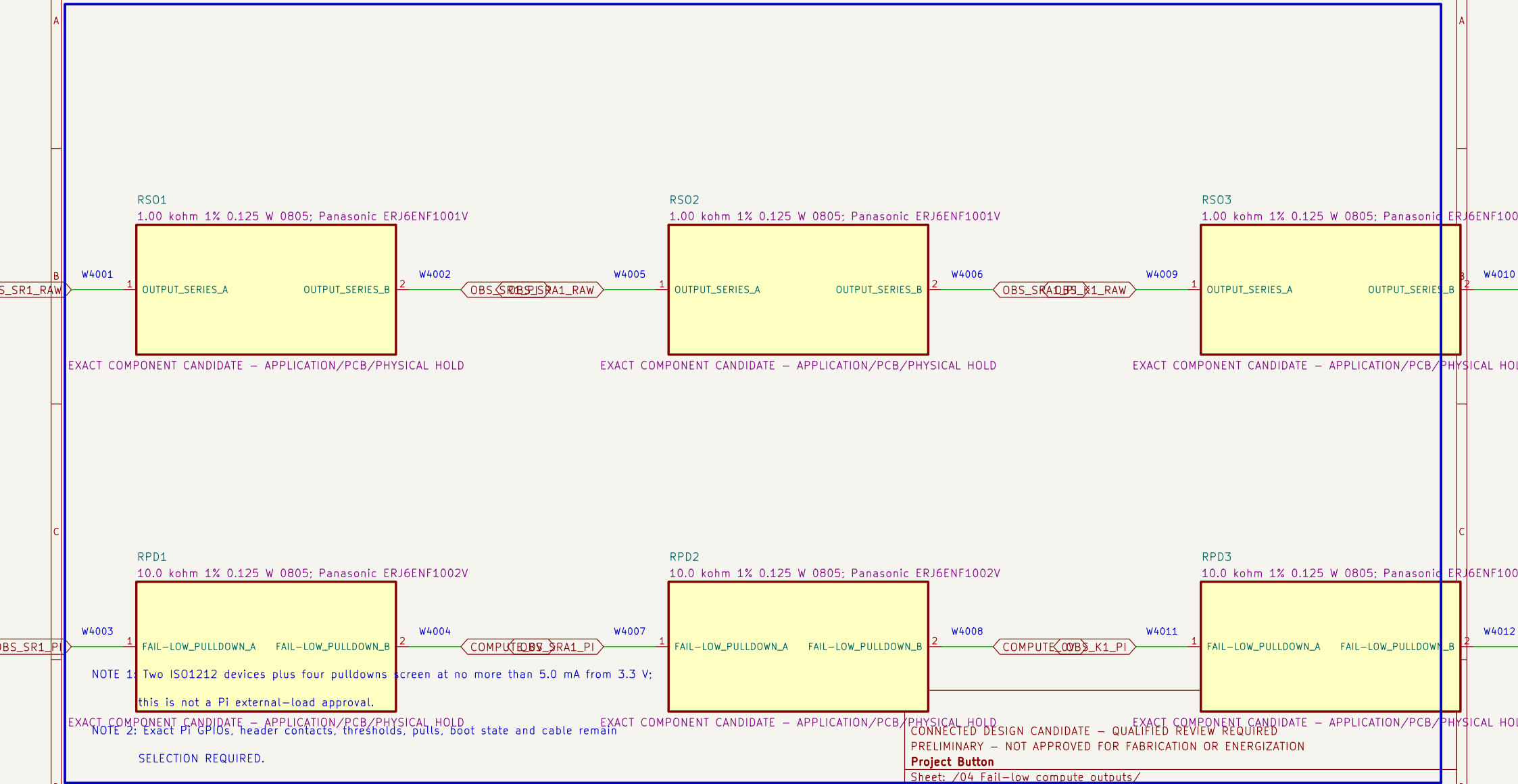
03 K1 and K2 diagnostic auxiliary inputs

2.70 kohm shunts raise current above Schneider signalling minima in the derived screen.



04 Fail-low compute outputs

Four 1 kohm series outputs and 10 kohm pulldowns; Pi pins remain selection required.



NOTE 1: Two ISO1212 devices plus four pulldowns screen at no more than 5.0 mA from 3.3 V;
this is not a Pi external-load approval.

NOTE 2: Exact Pi GPIOs, header contacts, thresholds, pulls, boot state and cable remain
SELECTION REQUIRED.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /04 Fail-low compute outputs/
File: 04_compute_outputs.kicad_sch
Title: PB HR-V0 R201 / P0.1 – 04
Size: A4 | Date: 2026-08-10 | Rev: R201 / P0.1
KiCad E.D.A. 10.0.5 | Id: 5/5